

PrivML: Test Outputs

Timestamp: 2025-09-19 20:20:45

Model ID: Model 47

Dataset Path: data/mnist/

Number of Images in Dataset: 10000

Training Size: 75%

Validation Size: 15%

Testing Size: 9%

Encrypted Testing Subset Size: 1%

Data Augmentation: True

Random Seed: 123

Plain Model Training

Epochs: 75

Batch Size: 16

Learning Rate: 0.0005

Optimizer: Adam

Loss Function: CrossEntropyLoss

Sample Output Count: 6

Plain Model Architecture

Hidden Layer Size: 128

Output Classes: 10

Conv Kernel Size: 7

Conv Stride: 3

Conv Padding: 0

Dropout Rate: 0.3

FHE Configuration

Encryption Scheme: CKKS

Scale (bits): 26

Polynomial Modulus Degree: 8192

Key Generation Level: 31

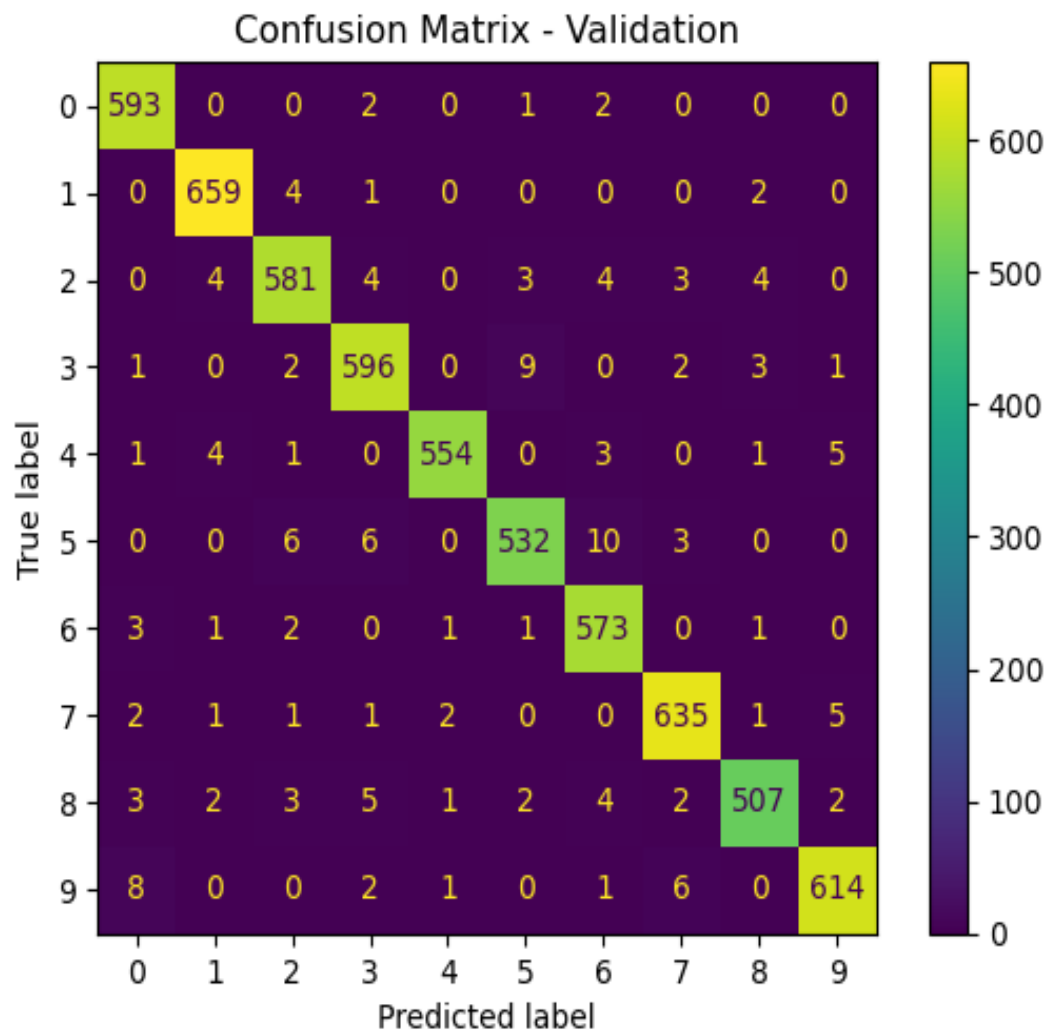
Decryption Stability Level: 31

Encrypted Loss Function: CrossEntropyLoss

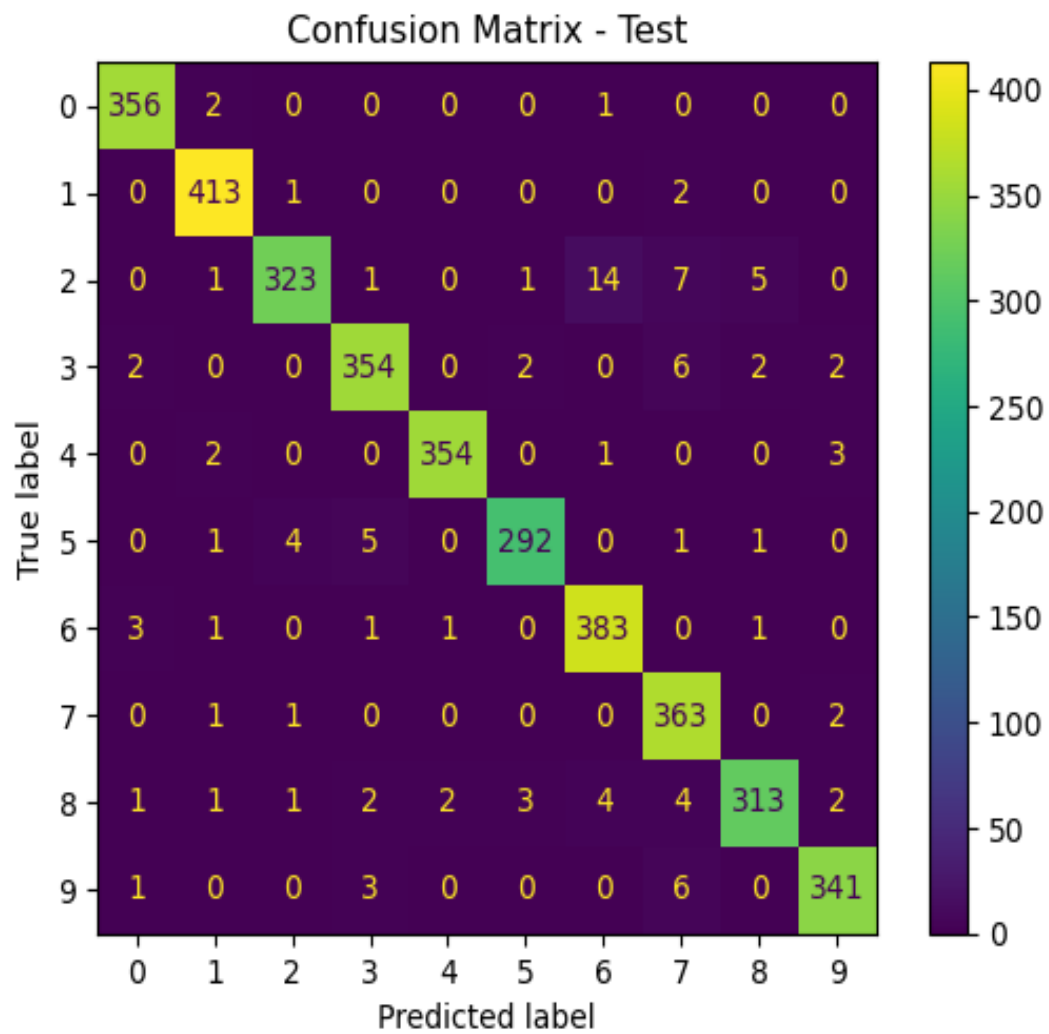
Coefficient Mod Bit Sizes: [31, 26, 26, 26, 26, 26, 26, 31]

Privacy-Preserving ML: ML Metrics Report

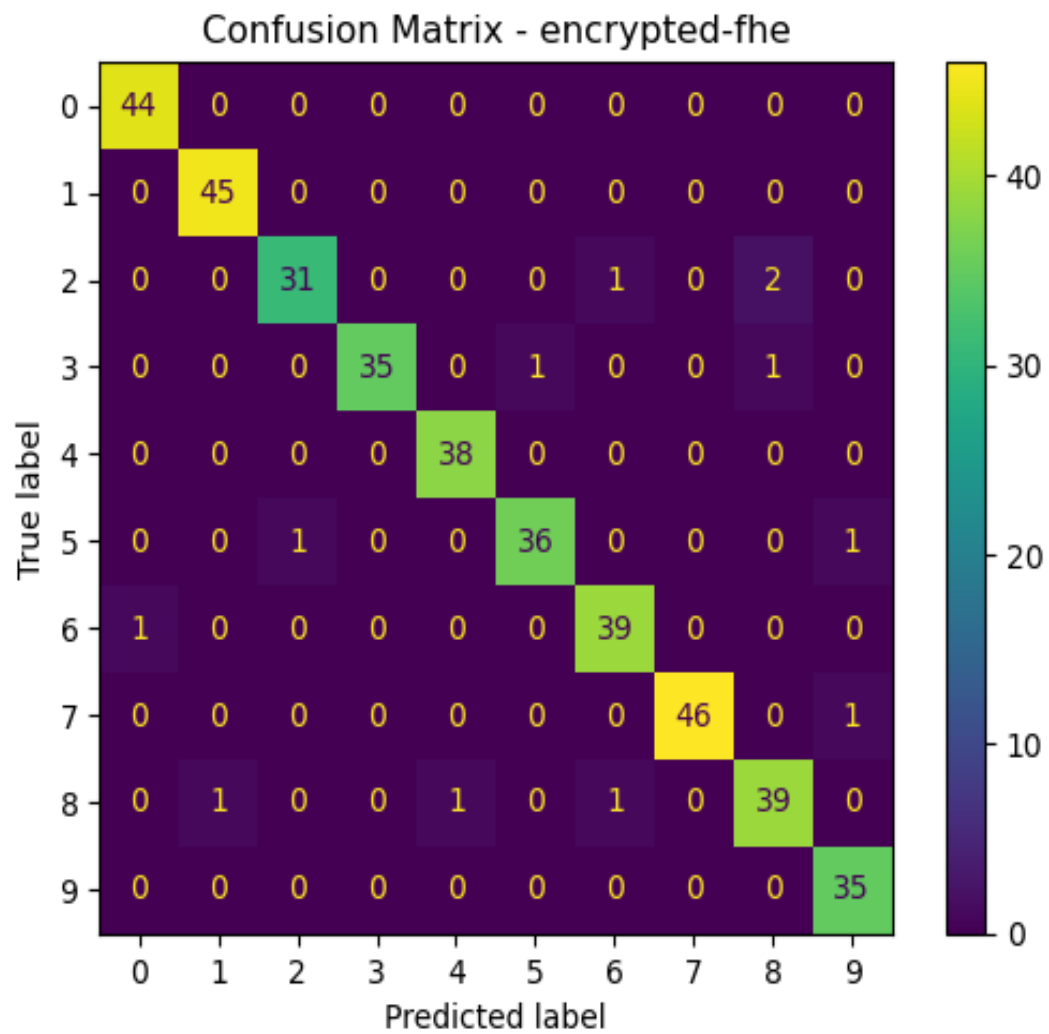
Confusion Matrix - Validation



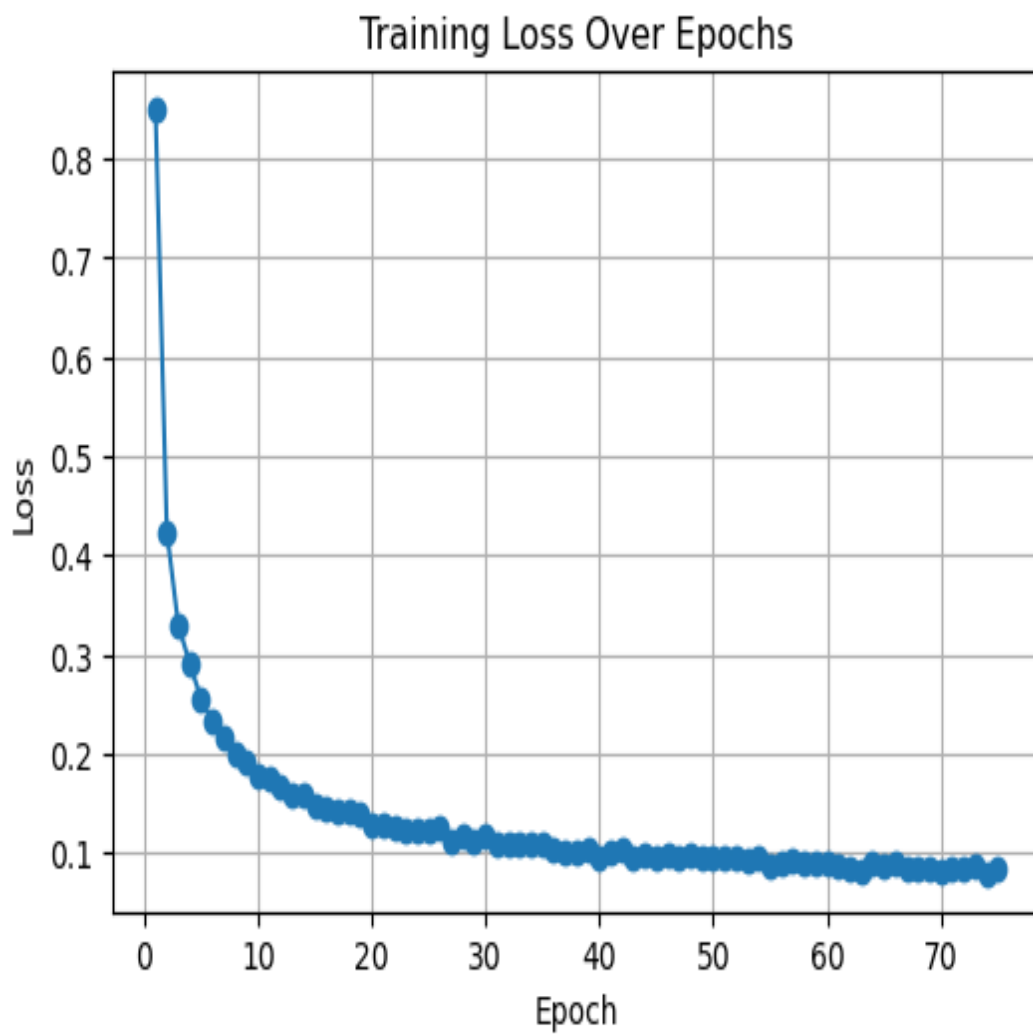
Confusion Matrix - Test



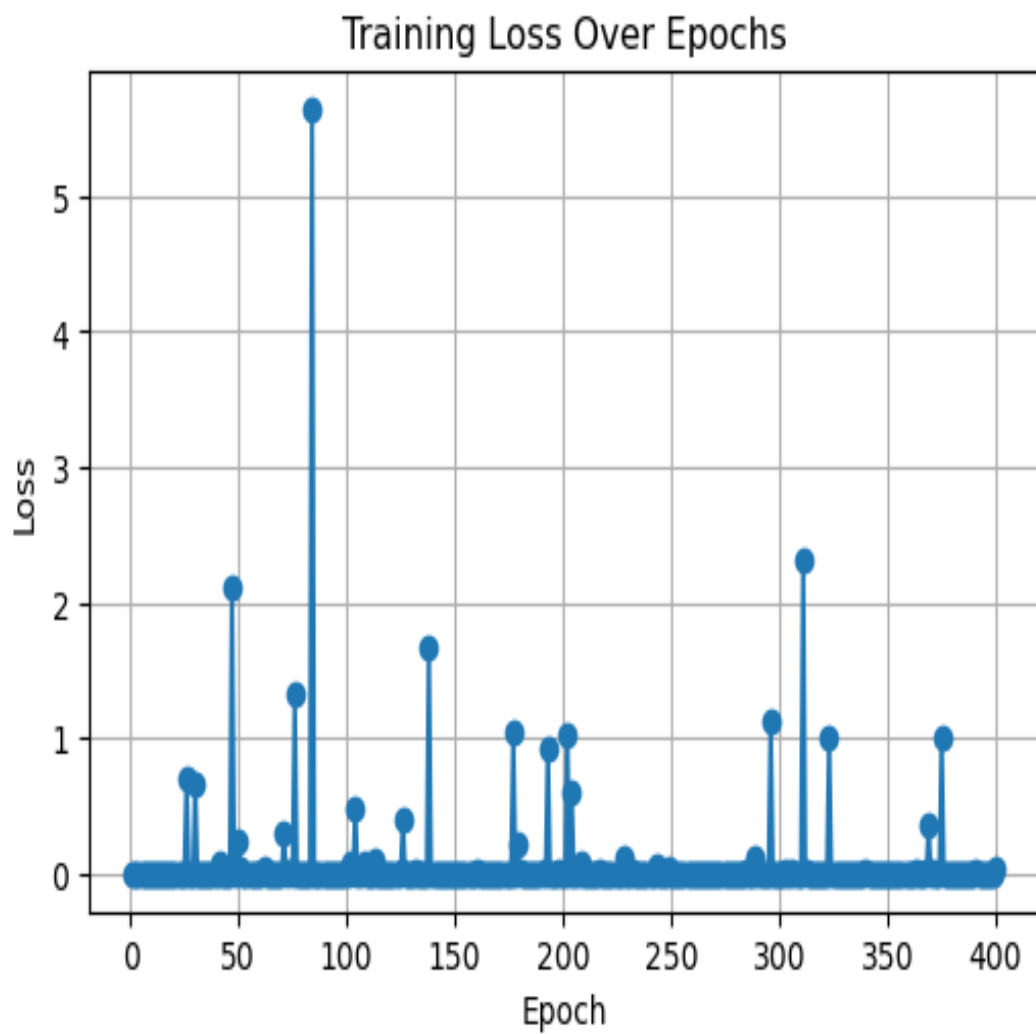
Confusion Matrix - encrypted-fhe



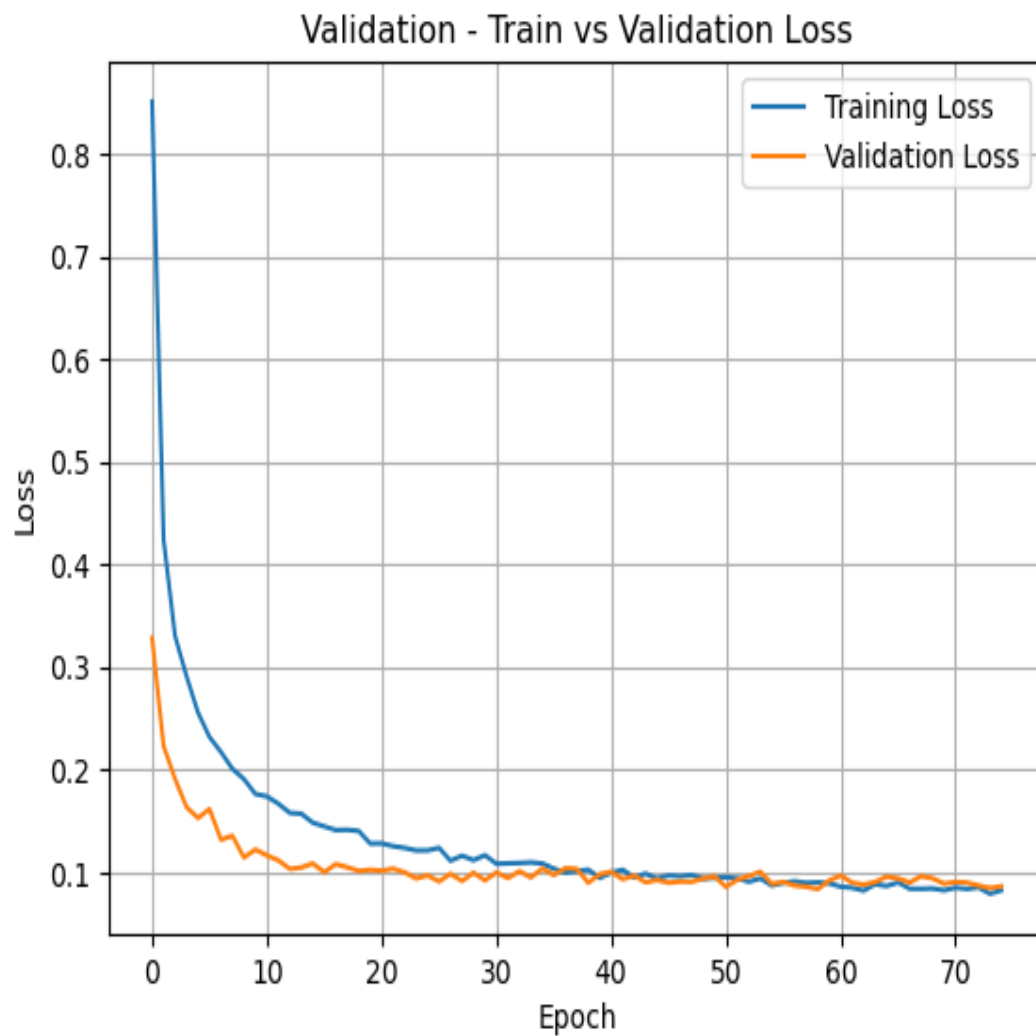
Training Loss - Validation



Training Loss - encrypted-fhe

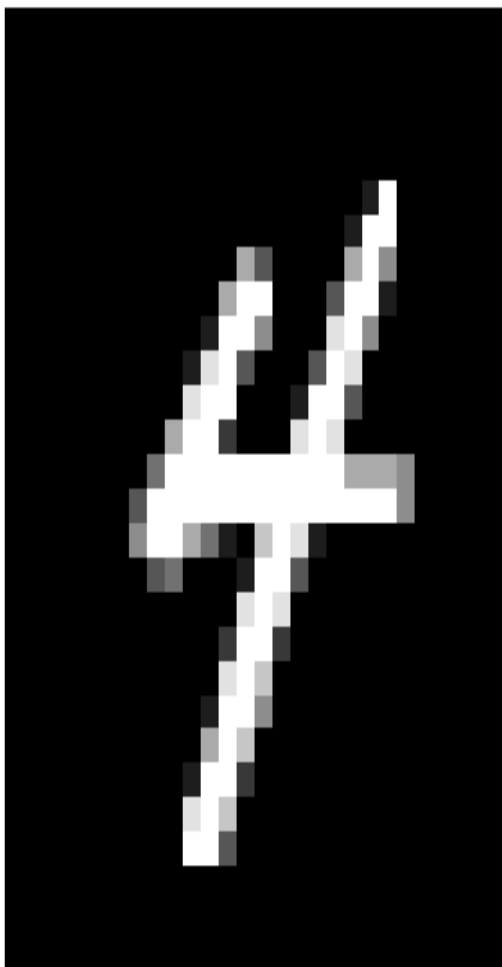


Train vs Validation Loss - Validation

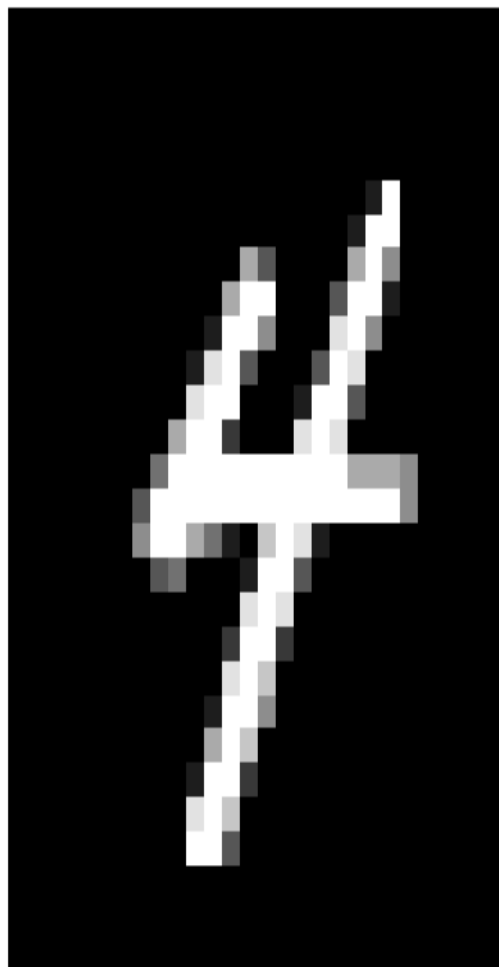


Predicted Images - Predicted Image

Ground Truth: 4

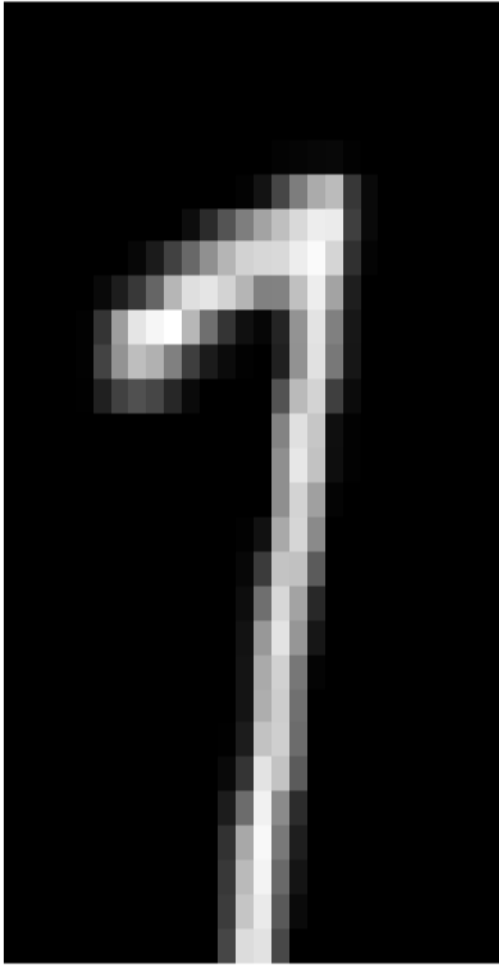


Predicted: 4

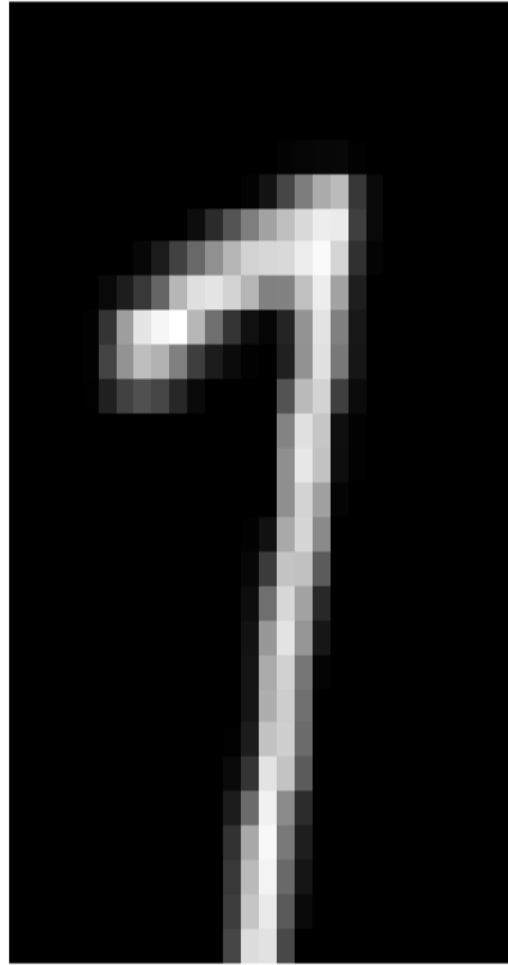


Predicted Images - Predicted Image

Ground Truth: 7

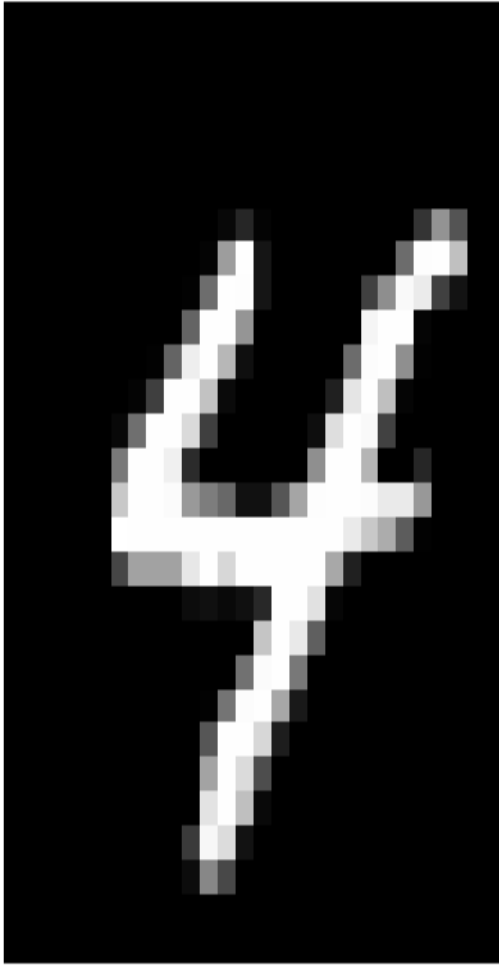


Predicted: 7

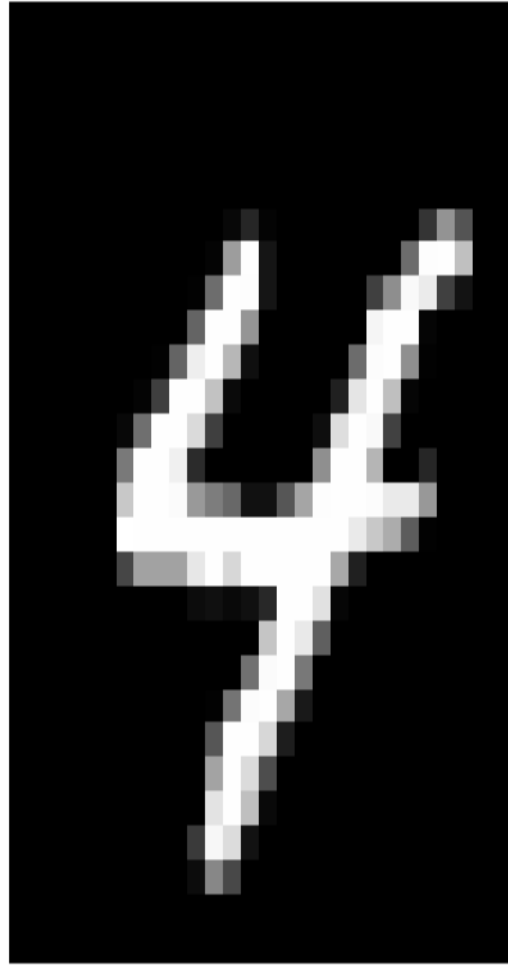


Predicted Images - Predicted Image

Ground Truth: 4

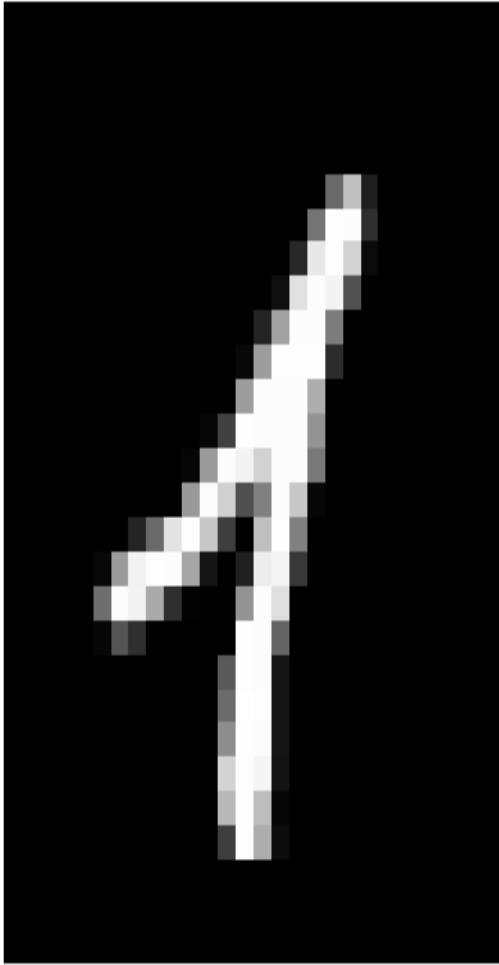


Predicted: 4

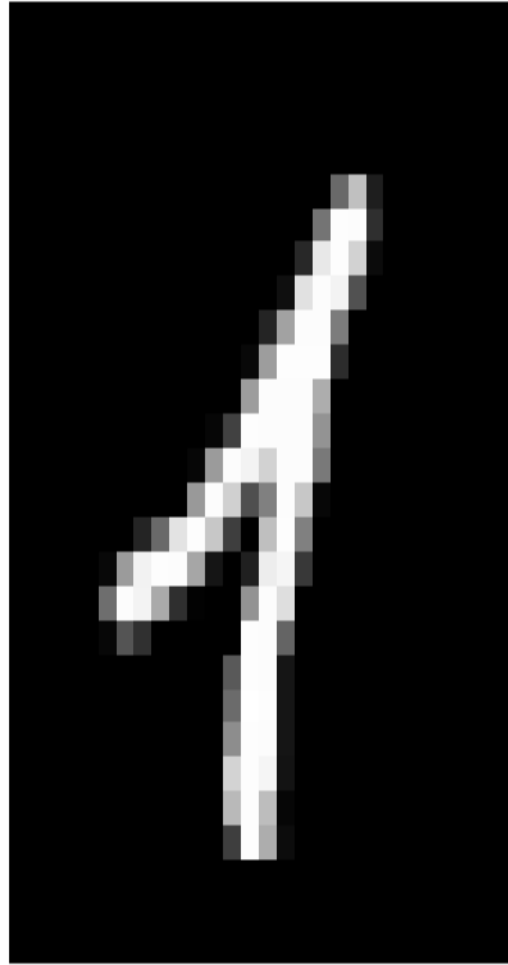


Predicted Images - Predicted Image

Ground Truth: 1

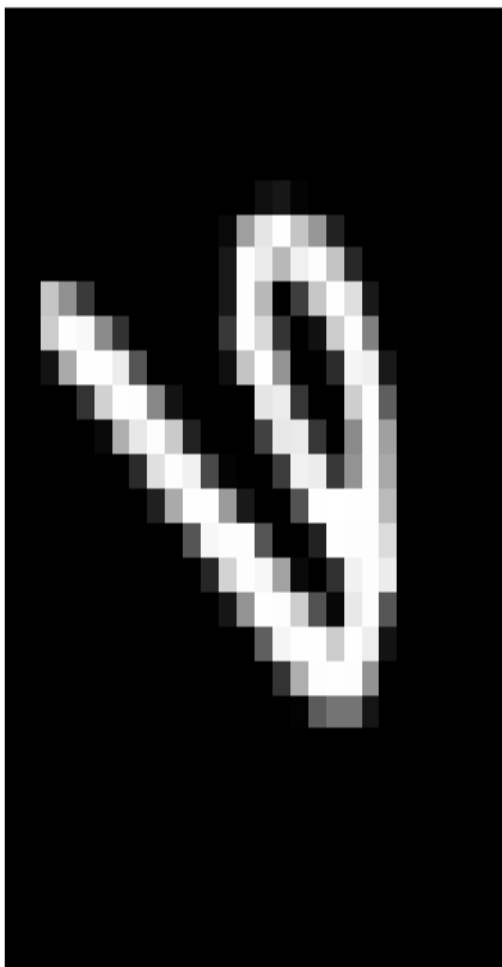


Predicted: 1

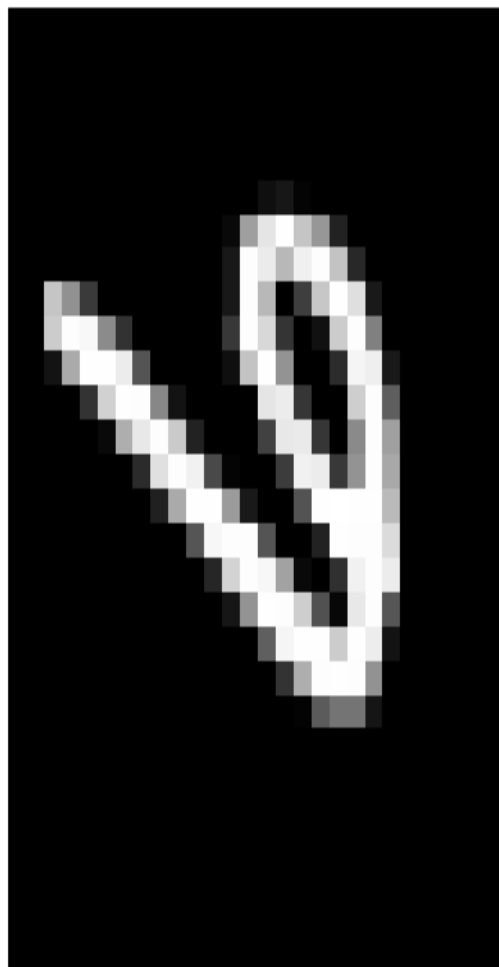


Predicted Images - Predicted Image

Ground Truth: 6

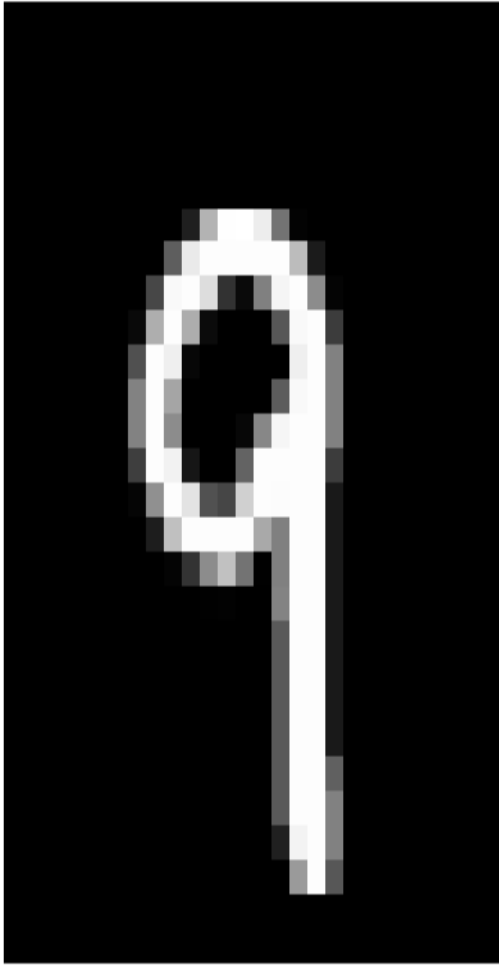


Predicted: 6

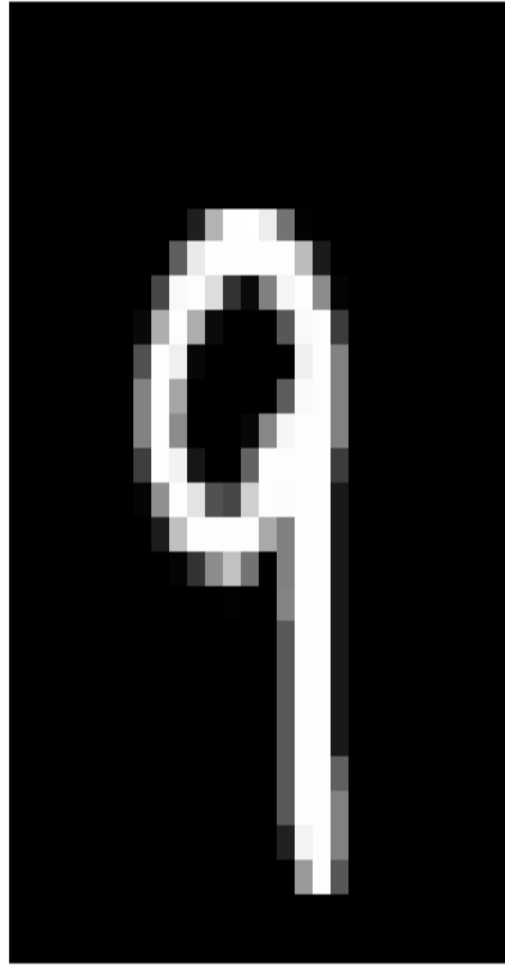


Predicted Images - Predicted Image

Ground Truth: 9



Predicted: 9



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit

encrypted-fhe: Python Version: 3.10.8

encrypted-fhe: Processor: arm

encrypted-fhe: CPU Usage (%): 10.6

encrypted-fhe: RAM Usage (Used GB): 8.57

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 63.1

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 5.7

encrypted-fhe: RAM Usage (Used GB): 8.35

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 62.2

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 8.6

encrypted-fhe: RAM Usage (Used GB): 8.36

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 62.2

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 11.1
encrypted-fhe: RAM Usage (Used GB): 8.27
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 61.9
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 8.2
encrypted-fhe: RAM Usage (Used GB): 8.31
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 62.4
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 8.7
encrypted-fhe: RAM Usage (Used GB): 8.33
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 62.5
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 7.7
encrypted-fhe: RAM Usage (Used GB): 8.03
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 63.7
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 8.5
encrypted-fhe: RAM Usage (Used GB): 8.04
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 63.9
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 9.9
encrypted-fhe: RAM Usage (Used GB): 7.92
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 63.9
encrypted-fhe: GPU: Not Available